

# Mathematics for Biomedical Engineering

L4:

## Matrices

Dr. Antonios Pouliopoulos  
*King's College London*

October 22nd, 2025



- to understand the basic mathematical representation of a matrix and its geometric interpretation
- to recognise the different forms of a matrix
- to perform operations, such as multiplication with matrices
- to understand how to compute determinants and inverses of matrices
- to understand the use of matrices in more complex mathematical and engineering problems

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices

# Introduction to matrices

A **matrix** is a rectangular array of numbers or expressions usually enclosed in brackets.

Matrices are often denoted by a capital letter.

$$A = \begin{pmatrix} 4 & 1 \\ 3 & 6 \\ -1 & 0 \end{pmatrix} \quad (1)$$

The **size** of a matrix is given by the number of rows  $\times$  the number of columns e.g.  $A$  is a  $3 \times 2$  matrix.

Individual numbers or expressions in a matrix are called the **elements**, usually denoted by a small letter e.g.  $a_{12}$  refers to the element in the first row, second column of  $A$ .

*Example: digital images can be thought of as large matrices.*

## Some special matrices

**Square Matrix** has the same number of columns as rows. The main **diagonal** is the diagonal running from top-left to bottom-right, containing the elements  $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ .

**Identity Matrix** is a square matrix with ones on the main diagonal and zeros elsewhere e.g.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (2)$$

**Transpose** of a matrix is obtained by switching the rows and columns, denoted by  $A^T$ .

*Example: Write-down the transpose of the matrix  $A$*

$$\begin{pmatrix} 6 & 1 & -3 & 0 \\ 9 & 2 & 3 & 7 \end{pmatrix} \quad (3)$$

## Indical notation of matrices

A matrix  $A$ , of size  $m \times n$  ( $m$  rows,  $n$  columns), may be represented by

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix} \quad (4)$$

An *individual entry*, or **element**, of matrix  $A$  from the  $i$ th row and the  $j$ th column can be represented by

$$(A)_{ij} = a_{ij} \quad (5)$$

Note that the transpose of a matrix can be written

$$(A^T)_{ij} = (A)_{ji} \quad (6)$$

## Trace of a matrix

The trace of a matrix is simply found by summing-up all of the entries along the diagonal of the matrix.

In indicial notation we can write this as

$$\text{Tr}(A) = \sum_{i=1}^n A_{ii} \quad (7)$$

*Example: Find the trace of the matrix*

$$\begin{pmatrix} 1 & 0 & 4 \\ 7 & -3 & 3 \\ 2 & 1 & -8 \end{pmatrix} \quad (8)$$



## Addition and subtraction of matrices

Matrices *of the same size* may be added to or subtracted from one another by simply adding or subtracting the corresponding elements.

### Addition and subtraction of matrices

Individual elements of  $A + B$  (i.e. the  $i$ th,  $j$ th entry) can be calculated by

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad (9)$$

### Transpose of the sum

The transpose of the sum of two matrices is the same as the sum of the individual matrices transposed

$$(A + B)^T = A^T + B^T \quad (10)$$

*Example: Prove the above relationship with matrices*

$$C = \begin{pmatrix} 7 & 2 \\ -2 & 1 \\ 4 & 3 \end{pmatrix} \quad D = \begin{pmatrix} -1 & 0 \\ 3 & 4 \\ 9 & 1 \end{pmatrix} \quad (11)$$

## Multiplication of a matrix by a number

Any matrix can be multiplied by a number. To do this, each element of the matrix is simply multiplied by the number.

### Scalar matrix multiplication

Individual elements of the matrix are multiplied by the scalar

$$(\lambda A)_{ij} = \lambda a_{ij}, \quad (12)$$

where  $\lambda$  is a scalar.

*Example: Find  $2A$  and  $\frac{1}{2}A$  if*

$$A = \begin{pmatrix} 3 & -1 & -1 \\ 0 & -2 & 4 \end{pmatrix} \quad (13)$$

### Scalar multiplication and transpose

$$(kA)^T = kA^T \quad (14)$$

*Example: Prove the above relationship with  $kA$  above.*

- 1 Introduction to matrices
- 2 Multiplication of matrices**
- 3 Determinants
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices

## Conditions required for matrix multiplication

If  $A$  is a  $m \times n$  matrix, and  $B$  a  $p \times q$  matrix,  $A$  and  $B$  can only be multiplied together if  $n = p$ .

### Condition for matrix multiplication

For the product of  $AB$  to exist, the number of columns of  $A$  must equal the number of rows of  $B$  i.e.  $n = p$ .

The result is another matrix whose size is  $m \times q$

### Result of matrix multiplication

$$\begin{matrix} A & B & = & C \\ m \times n & p \times q & & m \times q \end{matrix} \quad (15)$$

where  $n = p$ .

If the  $n \neq p$  then the product  $AB$  does not exist.

## Order of matrix multiplication

The *order* in which matrices are multiplied dictates the final result.

For example, if  $A$  is a  $3 \times 2$  matrix and  $B$  a  $2 \times 4$  matrix, then the product  $AB$  can be found, but the product  $BA$  does not exist.

### Matrix non-commutivity

Matrix multiplication is **not commutative** (i.e. the order is crucial).

In the product  $AB$  we say that  $A$  **premultiplies**  $B$ ; alternatively, we can say that  $B$  **postmultiplies**  $A$ .

If  $P$ ,  $Q$  and  $R$  are three matrices, then to calculate  $PQR$  we can do *either*  $(PQ)R$  or  $P(QR)$  - the final result is identical.

### Matrix associativity

Matrix multiplication is **associative**.

## How to multiply two matrices

Matrix multiplication is most easily explained and understood by considering indicial notation.

### Matrix multiplication

The element  $i, j$  of the product of  $AB$  is

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad (16)$$

where  $A$  is an  $m \times n$  matrix, and  $B$  is a  $p \times q$  matrix where  $n = p$ .

Therefore, if  $AB = C$  then the element  $c_{ij}$  is found from *row*  $i$  of  $A$  and *column*  $j$  of  $B$ .

*Example: Find the product of the two matrices*

$$\begin{pmatrix} 3 & 1 & 4 \\ 5 & 2 & -2 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ -6 & 4 \end{pmatrix} \quad (17)$$

# How to multiply two matrices

For the two matrices

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}, \quad C = AB = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \quad (18)$$

the product  $C = AB$  is found from (note:  $C$  is a  $2 \times 2$  matrix)

$$c_{11} = \sum_{k=1}^3 a_{1k} b_{k1} = \underline{a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}} \quad (19)$$

$$c_{12} = \sum_{k=1}^3 a_{1k} b_{k2} = a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \quad (20)$$

$$c_{21} = \sum_{k=1}^3 a_{2k} b_{k1} = a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} \quad (21)$$

$$c_{22} = \sum_{k=1}^3 a_{2k} b_{k2} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \quad (22)$$

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants**
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices



# Determinants

The **determinant** of a matrix is a very important quantity in mathematics, engineering and physics.

Some important facts about determinants:

- Depending on the size of the original matrix, the determinant may be a scalar, or it may be another matrix.
- The determinant of a matrix  $A$  is denoted by either  $\det(A)$  or  $|A|$ .
- A determinant can only be evaluated for a **square matrix** i.e. a  $2 \times 2$ ,  $3 \times 3$ ,  $4 \times 4$ ... etc. [You will only have to know up to and including  $3 \times 3$ .]
- The determinant of a  $1 \times 1$  'matrix' is just the number (i.e.  $|3| = 3$ ).
- The determinant of the transpose of a matrix is the same as the determinant of the original matrix i.e.  $|A^T| = |A|$ .
- A matrix that has zero determinant is said to be a **singular matrix**.

## Determinant of a $2 \times 2$ matrix

The determinant of a  $2 \times 2$  matrix is a *scalar* quantity, evaluated by a specific formula involving the elements of the matrix.

### Determinant of a $2 \times 2$ matrix

For a  $2 \times 2$  matrix  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  the determinant is defined by

$$\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad (23)$$

This formula should be learnt and memorised!

*Example: Compute the determinant of the matrix  $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$ . Now compute  $|A^T|$ .*

*Example: Under what conditions is the following matrix singular*

$$\begin{pmatrix} x & 3 \\ 2x & y \end{pmatrix} \quad (24)$$

## Minors and cofactors of a $3 \times 3$ matrix

Consider a  $3 \times 3$  matrix  $A$  where

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad (25)$$

If we chose an element of  $A$ , say  $a_{ij}$ , and *cross-out* the corresponding row and column i.e. the  $i$ th row and the  $j$ th column, we are left with a  $2 \times 2$  matrix.

The *determinant* of this  $2 \times 2$  matrix is called the **minor** of  $a_{ij}$ .

The minor of matrix  $a_{11}$  associated with matrix  $A$  is

$$\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} \quad (26)$$

This is obtained by eliminating all elements of row 1 and column 1.

*Example: What is the minor of  $a_{22}$  and  $a_{31}$ ?*

## Minors and cofactors of a $3 \times 3$ matrix

Cofactors are closely related to minors.

### Cofactors

The **cofactor** of  $a_{ij} = (-1)^{i+j} \times$  minor of  $a_{ij}$ .

The signs of the cofactors associated with each element (referred to as **place signs**) are best remembered by visualising them in a grid

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix} \quad (27)$$

If the place sign is  $+$ , the minor and cofactor are the same; if the place sign is negative, the cofactor is  $-1 \times$  minor.

*Example: Calculate the cofactors of elements 3, 9 and 2 in the matrix*

$$\begin{pmatrix} 3 & -1 & 6 \\ 9 & -5 & 2 \\ 0 & 4 & 7 \end{pmatrix} \quad (28)$$

## Determinant of a $3 \times 3$ matrix

The determinant of a  $3 \times 3$  matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad (29)$$

is found using

### Determinant of a $3 \times 3$ matrix

$$|A| = a_{11} \times (\text{its cofactor}) + a_{12} \times (\text{its cofactor}) + a_{13} \times (\text{its cofactor}) \quad (30)$$

This is called expanding along the first row. Note that the determinant can also be found by expanding along any row or column.

*Example: Calculate the determinant of the matrix*

$$A = \begin{pmatrix} 3 & -1 & 6 \\ 9 & -5 & 2 \\ 0 & 4 & 7 \end{pmatrix} \quad (31)$$

*Example: Calculate the determinant again, this time expanding along the second column.*

## Determinant of a $4 \times 4$ matrix

The determinant of a  $4 \times 4$  matrix is evaluated in a similar way to the previous case of the  $3 \times 3$  matrix.

The place signs in this case are

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix} \quad (32)$$

Expanding along the first row gives

$$|A| = a_{11} \times (\text{its cofactor}) + a_{12} \times (\text{its cofactor}) + a_{13} \times (\text{its cofactor}) + a_{14} \times (\text{its cofactor}) \quad (33)$$

Note that here, each of the cofactors will involve computing the determinant of a  $3 \times 3$ .

*Example: Find  $\det(A)$  by expanding along the first row*

$$A = \begin{pmatrix} 3 & -2 & 4 & 1 \\ 6 & 2 & 0 & 3 \\ 9 & 1 & 1 & -1 \\ 4 & 2 & -2 & 5 \end{pmatrix} \quad (34)$$

## Properties of determinants

P1 If  $A$  and  $B$  are two square matrices such that  $AB$  exists then

$$\det(AB) = \det(A) \det(B) \quad (35)$$

P2 If two rows or two columns of a matrix are equal, the determinant of the matrix is zero.

P3 Adding or subtracting a multiple of one row (or column) to/from another row (or column) leaves the determinant unchanged.

P4 If  $B$  is obtained from  $A$  by interchanging *two* rows or *two* columns then

$$\det(B) = -\det(A) \quad (36)$$

P5 If a square  $n \times n$  matrix  $A$  is multiplied by a scalar  $k$ , then

$$\det(kA) = k^n \det(A) \quad (37)$$

We can often use these properties to simplify the computation of determinants.

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants
- 4 Inverse of a matrix**
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices



## Definition

We represent the **inverse** of a square matrix  $A$  by  $A^{-1}$ .

### Definition of the inverse of a matrix

Multiplying a square matrix  $A$  by its inverse  $A^{-1}$  yields

$$AA^{-1} = A^{-1}A = I \quad (38)$$

where  $I$  is the identity matrix.

- Only *square* matrices have an inverse.
- If  $|A| = 0$ , then  $A$  does not have an inverse.
- If  $|A| \neq 0$ , then  $A$  does have an inverse.

Finding the inverse of a  $2 \times 2$  matrixInverse of a  $2 \times 2$  matrix

If  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  then

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad (39)$$

where  $|A| = ad - bc$

*Example: Find the inverse of*

$$B = \begin{pmatrix} 3 & -2 \\ 4 & -2 \end{pmatrix} \quad (40)$$

*Check that the product of  $B$  and  $B^{-1}$  gives  $I$ .*

## Finding the inverse of a matrix - the general case

To find the inverse of a general  $n \times n$  matrix (assuming that the inverse does exist and that  $|A| \neq 0$ ) we proceed thus:

### Inverse of an $n \times n$ matrix

- ① Find the transpose of  $A$  by interchanging the rows and columns of  $A$ .
- ② Replace each element of  $A^T$  by its cofactor. The resulting matrix is known as the **adjoint** of  $A$ , written  $\text{adj}(A)$ .
- ③ The inverse of  $A$  is then given by

$$A^{-1} = \frac{\text{adj}(A)}{|A|} \quad (41)$$

*Example: Find the inverse of the matrix*

$$\begin{pmatrix} 3 & 1 & 0 \\ 5 & 2 & -1 \\ 1 & 4 & -2 \end{pmatrix} \quad (42)$$

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)**
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices

## Representation of points using matrices

Matrices have a number of important uses when considering transformations of points or shapes in 2- and 3-dimensions. For example, transformations including scaling, translation, shearing, rotation and reflection can all be defined in terms of matrices.

A point is represented in 2-D by its coordinates,  $x$  and  $y$ , which may be written as a  $2 \times 1$  or a  $1 \times 2$  matrix i.e.  $(x \ y)$  or  $\begin{pmatrix} x \\ y \end{pmatrix}$ . [Here, we will use the latter form.]

Such matrices are referred to as **position matrices** or **position vectors**.

## Matrices as transformations

Mathematically, a transformation such as scaling, translation, shearing, rotation and reflection can be represented by a **transformation matrix**.

Since every figure/image/shape is made-up of discrete points, consider the action of a transformation matrix  $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  upon the point  $X = \begin{pmatrix} x \\ y \end{pmatrix}$

The matrix  $T$  *premultiplies* the point  $X$ , giving a new resultant  $X^*$ .

$$TX = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} = X^* \quad (43)$$

We can think of the point  $X$  being transformed into the point  $X^*$  by the matrix  $T$ .

*Find the new position of the point  $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$  transformed by the matrix  $\begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$*

## Transformation of lines

If a transformation matrix  $T$  (with *constant* elements) is applied to a straight line, then every point on the line gets transformed to a new location, forming a new straight line.

To find the transformation of a straight line we need only find the transformation of the two end-points and join these up.



**Figure:** The transformation of a straight-line is another straight-line.

# Scaling

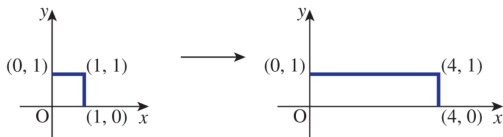
**Scaling** is performed using a *diagonal* matrix

$$\begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} \quad (44)$$

where the horizontal lengths are scaled by a factor of  $\alpha$  and the vertical lengths are scaled by a factor of  $\beta$ .

If  $\alpha > 1$  we have horizontal expansion; if  $\alpha < 1$  we have horizontal compression. Similarly for vertical expansion/ compression.

*Example:*

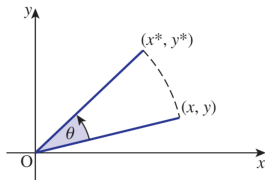


**Figure:** An example of horizontal scaling.



## Rotation

Rotation is an important and commonly used transformation. By convention, anticlockwise movement is considered to be positive.



**Figure:** The point  $(x, y)$  is rotated anticlockwise through an angle of  $\theta$  to the new point  $(x^*, y^*)$ .

### Rotation matrix

The transformation matrix  $T$  that rotates a point anticlockwise about the origin by an angle  $\theta$  is given by

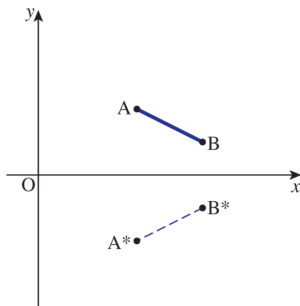
$$T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (45)$$

## Reflection in the $x$ -axis

### Reflection in the $x$ -axis

On reflection in the  $x$ -axis, the  $x$ -coordinate remains unchanged and the  $y$ -coordinate changes sign:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (46)$$



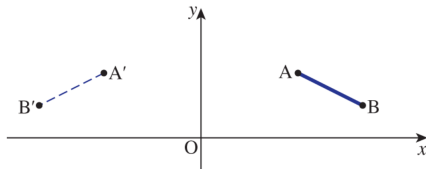
**Figure:** The line  $AB$  is reflected in the  $x$ -axis to produce the line  $A^*B^*$ .

## Reflection in the y-axis

### Reflection in the y-axis

On reflection in the y-axis, the y-coordinate remains unchanged and the x-coordinate changes sign:

$$T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad (47)$$



**Figure:** The line  $AB$  is reflected in the y-axis to produce the line  $A'B'$ .

## Reflection in the line $y = x$

### Reflection in the line $y = x$

Upon reflection in the line  $y = x$ , the  $x$ - and  $y$ -coordinate are interchanged:

$$T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (48)$$

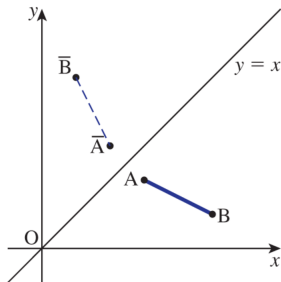


Figure:  $AB$  is reflected in the line  $y = x$  to produce  $\bar{A}\bar{B}$ .

## Reflection in the line $y = -x$

### Reflection in the line $y = -x$

Upon reflection in the line  $y = -x$ , the  $x$ - and  $y$ -coordinate are interchanged and also changed in sign:

$$T = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \quad (49)$$

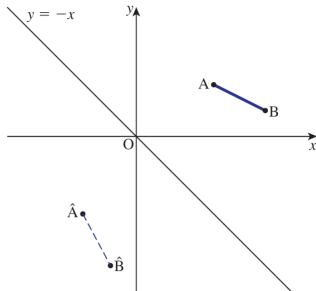


Figure:  $AB$  is reflected in the line  $y = x$  to produce  $\hat{A}\hat{B}$ .

## Shearing

A **shear** is a distortion. We can think, separately, of an  $x$ -direction shear and a  $y$ -direction shear, as shown below.



Figure: (left) An  $x$ -direction shear. (right) A  $y$ -direction shear.

### Shear in the $x$ -direction

$$T = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \quad (50)$$

which produces new points  $x' = x + ky$  and  $y' = y$ .

### Shear in the $y$ -direction

$$T = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \quad (51)$$

which produces new points  $x' = x$  and  $y' = y + kx$ .

# Translation

A **translation** is a movement in a specific direction by a specific amount.

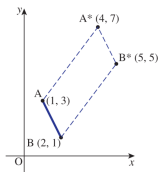


Figure:  $AB$  is translated to  $A^*B^*$ .

The previous transformations have all been represented by *matrix multiplication*; translations are represented by *matrix addition*.

In order to be able to represent them as multiplications, however, we introduce **homogeneous coordinates**

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad (52)$$

# Translation

## Translation

If the  $x$ -translation is  $t_x$  and the  $y$ -translation is  $t_y$

$$T = \begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix} \quad (53)$$

Apply this to the point  $(x, y)$  (in homogeneous coordinates) gives

$$\begin{pmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x + t_x \\ y + t_y \\ 1 \end{pmatrix} \quad (54)$$

which gives us our mapped point as

$$\begin{pmatrix} x + t_x \\ y + t_y \end{pmatrix} \quad (55)$$

*Example: What is the translation matrix associated with the Figure above?*



## Composite transformations

A series of different transformations is represented by a *product of matrices*. Care must be taken, though, as matrix multiplication is not commutative ( $AB \neq BA$ ).

A sequence of transformation is called a **composite transformation**.

To allow translation (which uses homogeneous coordinates) and other translations to work together, we now change all the previous translations to use homogeneous coordinates as well

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow T = \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (56)$$

*Example: The point (4,1) is reflected in the line  $y = x$  and then rotated anticlockwise about the origin by 70 degrees. Calculate the composite transformation matrix and hence the final position of the point.*

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors**
- 7 Applications & uses of matrices

## Trivial & non-trivial solutions

Consider the simultaneous equations

$$ax + by = 0 \quad (57)$$

$$cx + dy = 0 \quad (58)$$

The **trivial solution** of this system is simply  $x = 0, y = 0$ .

The **non-trivial solution** of this system is all other solutions for  $x$  and  $y$  that still satisfy the above system, but is *not* the trivial solution.

### Conditions for non-trivial solutions

For the system

$$AX = 0 \quad (59)$$

- If  $\det(A) = 0$ , the system has non-trivial solutions.
- If  $\det(A) \neq 0$ , the system has only the trivial solutions.

*Example: does the system below have non-trivial solutions?*

$$x + 2y = 0 \quad (60)$$

$$3x + 6y = 0 \quad (61)$$

# Eigenvalues

Some matrices have special vectors associated with them, such that when those vectors are pre-multiplied by the matrix, the result is *exactly the same matrix multiplied by a scalar*

$$AX = \lambda X \quad (62)$$

From this we can deduce that

$$AX - \lambda IX = \mathbf{0} \quad \Rightarrow \quad (A - \lambda I)X = \mathbf{0} \quad (63)$$

For the above equation to have *non-trivial* solutions (i.e. to actually have one of these special vectors,  $X$ ) requires that

$$|A - \lambda I| = 0 \quad (64)$$

# Eigenvalues

## Eigenvalues

The **characteristic equation** of a square matrix  $A$  is given by

$$|A - \lambda I| = 0 \quad (65)$$

Solutions of this equation are the **eigenvalues** of  $A$ . These are the values of  $\lambda$  for which  $AX = \lambda X$  has non-trivial solutions.

*Example: Find the eigenvalues associated with the matrix*

$$\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \quad (66)$$

# Eigenvectors

For each eigenvalue there is a non-trivial solution of the system - this solution is called an **eigenvector**.

To find the (normalised) eigenvectors associated with a specific eigenvalue of a matrix:

- 1 substitute in the specific value of  $\lambda$
- 2 expand-out the linear system (i.e. get equations in terms of  $x$  and  $y$ )
- 3 solve the equations to find  $x$  in terms of  $y$
- 4 set  $x = 1$  and obtain the value of  $y$
- 5 normalise the eigenvector

*Example: Find the eigenvectors associated with the matrix*

$$\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \quad (67)$$

- 1 Introduction to matrices
- 2 Multiplication of matrices
- 3 Determinants
- 4 Inverse of a matrix
- 5 Geometrical transformations (Computer Graphics)
- 6 Eigenvalues & Eigenvectors
- 7 Applications & uses of matrices**

## Systems of linear equations

Matrices can be used to solve **systems of linear equations**. For example, the system of linear equations

$$a_1x + b_1y = k_1 \quad (68)$$

$$a_2x + b_2y = k_2 \quad (69)$$

can be rewritten as the matrix equation

$$AX = B \quad (70)$$

where

$$A = \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix}, \quad B = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} \quad (71)$$

Our goal is to find the elements of  $X$  (the vector of unknowns).

The matrix  $A$  and the vector  $B$  simply contain numbers.



## Solving equations using the inverse matrix method

To solve the previous matrix equation

$$AX = B \quad (72)$$

If we can find the *inverse* of matrix  $A$  we can then pre-multiply both sides by  $A^{-1}$

$$A^{-1}AX = A^{-1}B \quad (73)$$

However, the left-hand side can be written as  $(A^{-1}A)X = IX = X$  giving

$$X = A^{-1}B \quad (74)$$

### Inverse solution method

Given  $AX = B$ , then  $X = A^{-1}B$  if  $A^{-1}$  exists.

*Example: Solve the follow set of simultaneous equations using the inverse matrix method.*

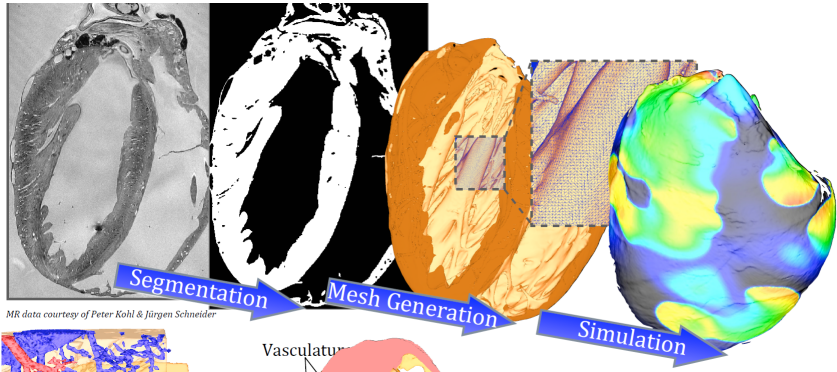
$$7x + 2y = 12 \quad (75)$$

$$3x + y = 5 \quad (76)$$

## Importance of matrices in numerical analysis

- In many aspects of applied maths, engineering, physics, etc, we end-up with a system of linear equations that need solving.
- This occurs when we *discretise* a large problem and just try to find solutions at specific points, or nodes, in space, using methods such as the finite element or finite difference method.
- These are often very large systems, sometimes with million of 'degrees of freedom'.
- In these cases, computing the inverse of a very large matrix is simply infeasible - instead, *elimination* or *iterative* methods are often used.

# Importance of matrices in numerical analysis



**Figure:** Discretisation of cardiac model to solve equations describing electrical activation of the heart.